

SAT Unit 4.4

Circle

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Overview

- 1 Arc Length and Area of a Sector
- 2 Properties of the Circle
- 3 Equation of a Circle in XY-Plane
- 4 Properties of the Circle

Arc Length and Area of a Sector

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Arc Length of a Circle

Arc Length Formula:

$$s = r\theta$$

(where s = arc length, r = radius, θ = central angle in radians)

Example: Find the arc length of a circle with:

- Radius: $r = 6$
- Central angle: $\theta = 60^\circ = \frac{\pi}{3}$ radians

$$s = r\theta = 6 \cdot \frac{\pi}{3} = 2\pi$$

SAT Tip: Do not use this formula unless θ is in radians!

Area of a Sector

Definition: A sector is a "pizza slice" portion of a circle.

Formula (Radians):

$$\text{Area} = \frac{1}{2}r^2\theta$$

Formula (Degrees):

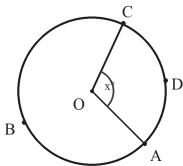
$$\text{Area} = \frac{\theta^\circ}{360^\circ} \cdot \pi r^2$$

Example: Find the area of a sector of a circle with radius 4 and angle 45° :

$$\text{Area} = \frac{45}{360} \cdot \pi \cdot 4^2 = \frac{1}{8} \cdot 16\pi = 2\pi$$

SAT Reminder: Be sure whether angle is in radians or degrees before using a formula.

Question: Arc Length and Central Angle



The circle above has center O , the length of arc $\widehat{ADC}$ is 5π , and $x = 100$. What is the length of arc $\widehat{ABC}$?

- A. 9π
- B. 13π
- C. 18π
- D. $\frac{13}{2}\pi$

Answer: Arc Length and Central Angle

We are told:

- Length of arc $\widehat{ADC} = 5\pi$
- Central angle $\angle AOC = x = 100^\circ$

Since the entire circle represents 360° , the arc $\widehat{ABC}$ must span:

$$360^\circ - 100^\circ = 260^\circ$$

Using the proportion of arc to full circle:

$$\frac{260}{100} = \frac{\text{Length of } \widehat{ABC}}{\text{Length of } \widehat{ADC}} = \frac{L}{5\pi} \Rightarrow L = 5\pi \cdot \frac{260}{100} = 13\pi$$

Correct Answer: B. 13π

Properties of the Circle

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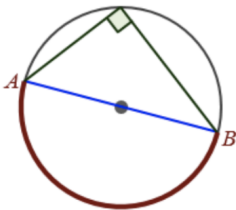
Angles Inscribed in a Semicircle

Theorem: Any angle inscribed in a semicircle is a **right angle (90°)**.

- If a triangle is inscribed in a circle and one side is the diameter, then the angle opposite that side is 90° .

Example: If triangle ACB is inscribed in a circle and AB is a diameter, then:

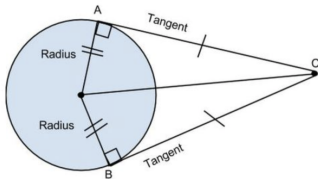
$$\angle C = 90^\circ$$



SAT Tip: Look for a triangle with one side as a diameter — it's a right triangle!

Radius and Tangent Line

Key Property: The radius of a circle is **perpendicular** to the tangent line at the point of contact.



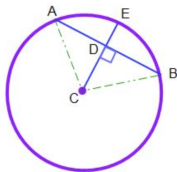
- Use this right angle in triangle questions.
- Often used in slope problems — if radius has slope m , tangent has slope $-\frac{1}{m}$.

SAT Tip: If the question mentions “tangent”, check for perpendicularity with radius.

Radius Bisecting a Chord

Theorem: If a radius (or diameter) is perpendicular to a chord, it **bisects** the chord.

If $AB \perp OE$, then $AD = DB$

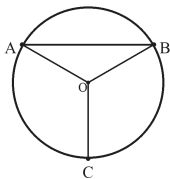


- Can be used to find missing lengths using right triangle properties.
- Also useful for coordinate geometry questions involving symmetry.

Note: The converse is also true — if a radius bisects a chord, it's perpendicular.

Useful for: Midpoint formula, Pythagorean theorem, and algebraic proofs.

Question: Arc Length in Circle Geometry



Point O is the center of the circle above, and the measure of $\angle OAB$ is 30° . If the length of $\overline{OC}$ is 18, what is the length of arc $\widehat{AB}$?

- A. 9π
- B. 12π
- C. 15π
- D. 18π

Answer: Arc Length in Circle Geometry

Since $\angle OAB = 30^\circ$, the central angle for arc AB is 60° (as it spans angle AOB). The radius is $r = 18$, so the circumference of the full circle is:

$$C = 2\pi r = 2\pi(18) = 36\pi.$$

The arc $\widehat{AB}$ represents $\frac{120^\circ}{360^\circ} = \frac{1}{3}$ of the full circle, so its length is:

$$\frac{1}{3} \times 36\pi = 12\pi.$$

Correct Answer: B. 12π

Equation of a Circle in XY-Plane

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Equation of a Circle (Coordinate Plane)

Standard Form:

$$(x - h)^2 + (y - k)^2 = r^2$$

Where:

- (h, k) is the center,
- r is the radius.

Example: Circle with center $(3, -2)$ and radius 5:

$$(x - 3)^2 + (y + 2)^2 = 25$$

SAT Skills:

- Completing the square to rewrite equations.
- Identify center and radius from expanded forms.

Completing the Square to Find Circle Equation

Given:

$$x^2 + y^2 - 6x + 8y + 9 = 0$$

Steps:

- ① Group x and y terms:

$$(x^2 - 6x) + (y^2 + 8y) = -9$$

- ② Complete the square:

$$(x - 3)^2 - 9 + (y + 4)^2 - 16 = -9$$

- ③ Simplify:

$$(x - 3)^2 + (y + 4)^2 = 16$$

Final Answer: Center = $(3, -4)$, Radius = 4

SAT Tip: Always isolate constants and complete square carefully on both x and y terms.

Question: Completing the Square

The graph of the equation

$$x^2 + x + y^2 + y = \frac{199}{2}$$

in the xy -plane is a circle. What is the length of the circle's radius?

Answer: Completing the Square – Radius of a Circle

To convert the given equation to the standard form of a circle, complete the square for both x and y :

$$x^2 + x + y^2 + y = \frac{199}{2}$$

Complete the square:

$$x^2 + x = \left(x + \frac{1}{2}\right)^2 - \frac{1}{4}, \quad y^2 + y = \left(y + \frac{1}{2}\right)^2 - \frac{1}{4}$$

So,

$$\left(x + \frac{1}{2}\right)^2 - \frac{1}{4} + \left(y + \frac{1}{2}\right)^2 - \frac{1}{4} = \frac{199}{2}$$

$$\left(x + \frac{1}{2}\right)^2 + \left(y + \frac{1}{2}\right)^2 = \frac{199}{2} + \frac{1}{2} = \frac{200}{2} = 100$$

Thus, the radius is:

Practices

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Equation of a Circle

Question: The center of a circle in the xy -plane is at $(1, -1)$ and the point $(-5, 7)$ lies on the circle. Which of the following equations represents this circle?

A $(x + 1)^2 + (y - 1)^2 = 10$

B $(x - 1)^2 + (y + 1)^2 = 10$

C $(x + 1)^2 + (y - 1)^2 = 100$

D $(x - 1)^2 + (y + 1)^2 = 100$

Solution

Solution: - The center is at $(1, -1)$. - Find the radius using the distance formula:

$$r = \sqrt{(-5 - 1)^2 + (7 - (-1))^2} = \sqrt{(-6)^2 + (8)^2} = \sqrt{36 + 64} = \sqrt{100} = 10$$

- The circle's equation is:

$$(x - 1)^2 + (y + 1)^2 = 100$$

Answer: D

Circles in the xy -Plane

Question: In the xy -plane, circle A is defined by the equation $(x - 3)^2 + (y - 4)^2 = 16$, and circle B is defined by the equation $(x - 3)^2 + (y - 4)^2 = n$, where n is a constant. The diameter of circle B is 48 units longer than the diameter of circle A . What is the value of n ?

Solution

Solution:

- Circle A :

$$r_A = \sqrt{16} = 4 \quad \Rightarrow \quad d_A = 2 \times 4 = 8$$

- Circle B :

$$d_B = d_A + 48 = 8 + 48 = 56 \quad \Rightarrow \quad r_B = 28$$

- Therefore:

$$r_B^2 = 28^2 = 784$$

Hence:

$$\boxed{n = 784}$$

Circle in the xy -Plane

Question: The given equation represents a circle in the xy -plane:

$$x^2 + y^2 + 7x + y = -\frac{7}{2}$$

What is the length of the radius of the circle?

Solution

- Start with:

$$x^2 + 7x + y^2 + y = -\frac{7}{2}$$

- Complete the square:

$$x^2 + 7x \rightarrow x^2 + 7x + \left(\frac{7}{2}\right)^2 = \left(x + \frac{7}{2}\right)^2 - \frac{49}{4}$$

$$y^2 + y \rightarrow y^2 + y + \left(\frac{1}{2}\right)^2 = \left(y + \frac{1}{2}\right)^2 - \frac{1}{4}$$

- Combine:

$$\left(x + \frac{7}{2}\right)^2 - \frac{49}{4} + \left(y + \frac{1}{2}\right)^2 - \frac{1}{4} = -\frac{7}{2}$$

- Bring constants to the right:

$$\begin{aligned}\left(x + \frac{7}{2}\right)^2 + \left(y + \frac{1}{2}\right)^2 &= -\frac{7}{2} + \frac{49}{4} + \frac{1}{4} \\ &= -\frac{7}{2} + \frac{50}{4} \\ &= -\frac{14}{4} + \frac{50}{4} = \frac{36}{4} = 9\end{aligned}$$

- Therefore:

$$r = \sqrt{9} = \boxed{3}$$

Triangle DEF

Question: In the xy -plane, a circle has center D with coordinates $(m, 16)$, where m is a constant. Points E and F lie on the circle. Point E lies on the circle and has coordinates $(m + 4\sqrt{10}, 34)$, and point F has coordinates $(n, 16)$, where n is a constant greater than m . What is the area, in square units, of triangle DEF ?

- A 198
- B $72\sqrt{10}$
- C 242
- D $99\sqrt{10}$

Solution

Solution:



198.

Tangent Line and Circle

Question: A circle in the xy -plane has the equation

$$(x - 2)^2 + (y - 2)^2 = 25.$$

Line m is tangent to this circle at the point $(5, 6)$. What is the y -coordinate of the y -intercept of line m ?

Solution

Solution:

- The center of the circle is at $(2, 2)$, and the radius is 5.
- The tangent line at point $(5, 6)$ is perpendicular to the radius at that point.
- The slope of the radius is:

$$\frac{6 - 2}{5 - 2} = \frac{4}{3}.$$

- Therefore, the slope of line m is the negative reciprocal:

$$-\frac{3}{4}.$$

- Using the point-slope form:

$$y - 6 = -\frac{3}{4}(x - 5).$$

- Setting $x = 0$ to find the y -intercept:

$$y = 6 + \frac{15}{4} = \frac{24}{4} + \frac{15}{4} = \frac{39}{4}.$$

- Therefore:

$$\boxed{\frac{39}{4}}.$$

Thank You!

We appreciate your time and focus.

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